SPLITTING OFF SEVERAL ISOLATED FIBRE POINTS BY A FINITE PART OVER AN ETALE NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12093-12112 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 12114-12129, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Separated morphisms, isolated points of a fibre, elementary etale neighbourhoods, and open and closed decompositions; the construction is retained.

Source reference. Original lemma 02LL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11974–11981. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01WN: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12129–12129. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W6: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12129–12129. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01KU: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12129–12129. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $s \in S$. Let $x_1, \ldots, x_n \in X_s$. Assume that*

(1) *f is locally of finite type,*
(2) *f is separated, and*
(3) *$x_1, \ldots, x_n$ are pairwise distinct isolated points of X_s.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Then there exists an elementary étale neighbourhood $(U, u) \to (S, s)$ *and a decomposition*

$$U \times_S X = W \amalg V_1 \amalg \ldots \amalg V_n$$

into open and closed subschemes such that the morphisms $V_i \to U$ *are finite, the fibres of* $V_i \to U$ *over* u *are singletons* $\{v_i\}$*, each* v_i *maps to* x_i *with* $\kappa(x_i) = \kappa(v_i)$*, and the fibre of* $W \to U$ *over* u *contains no points mapping to any of the* x_i*.*

Proof. Choose $(U, u) \to (S, s)$ and $V_i \subset X_U$ as in Lemma 02LL. Since $X_U \to U$ is separated (Schemes, Lemma 01KU) and $V_i \to U$ is finite hence proper (Morphisms, Lemma 01WN) we see that $V_i \subset X_U$ is closed by Morphisms, Lemma 01W6. Hence $V_i \cap V_j$ is a closed subset of V_i which does not contain v_i. Hence the image of $V_i \cap V_j$ in U is a closed set (because $V_i \to U$ proper) not containing u. After shrinking U we may therefore assume that $V_i \cap V_j = \emptyset$ for all i, j. This gives the decomposition as in the lemma. □